

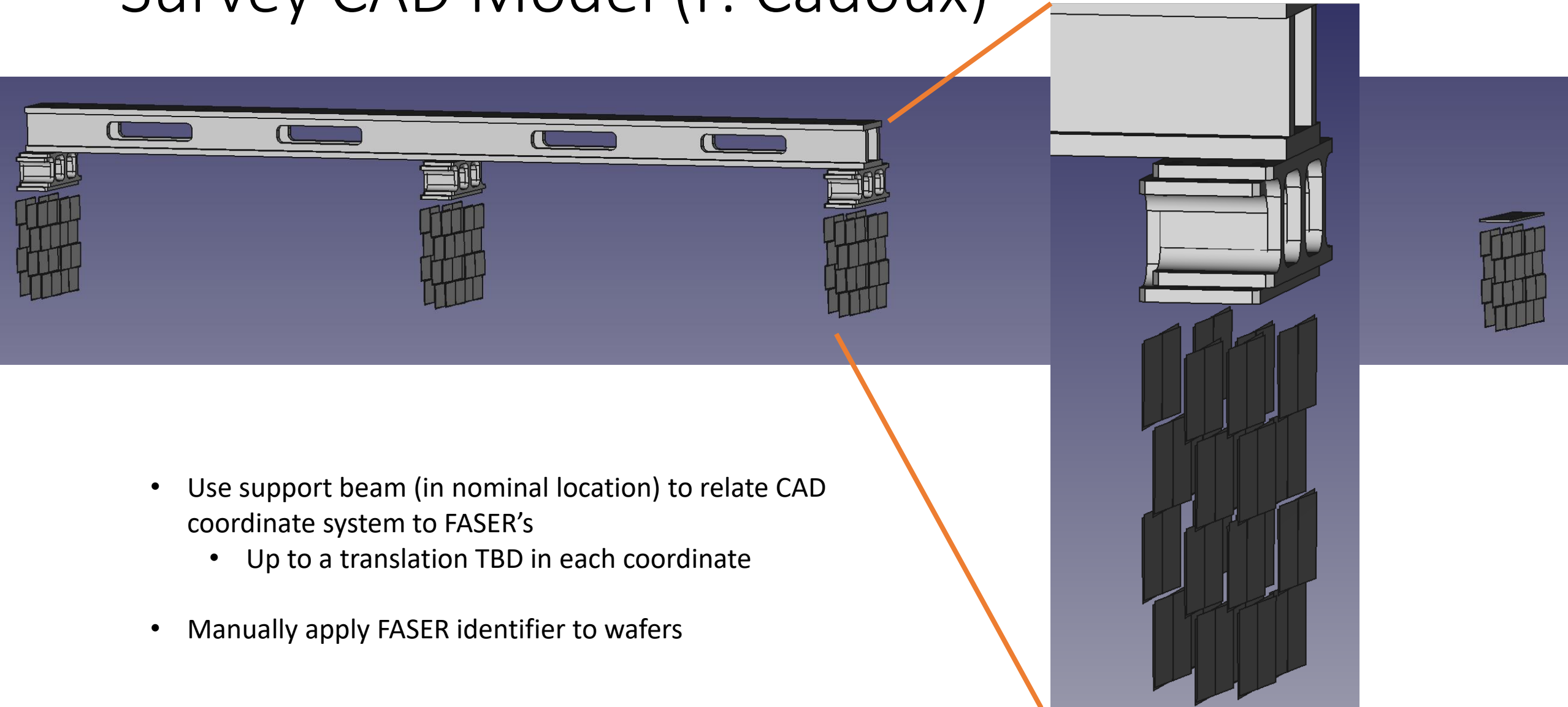
Comparing Tracker Survey with Nominal Geometry

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Survey CAD Model (F. Cadoux)



Wafer Separation

- In the CAD file, the wafers of each module are about 1.3 mm apart
- ATLAS quotes the nominal module thickness in the sensor region as 1.15 mm
 - Each sensor is 0.285 mm thick
 - The sensor centers must therefore be ~ 0.58 mm apart
 - The ATLAS geometry description (and ours) uses 0.6 mm for the separation
- Unclear how to interpret the z positions in the CAD file(?)
 - For all plots and statistics, I will only use the front sensor (pigtail side)
 - This is the one we align
 - There may still be some related anomalies (see later slides)

Nominal Geometry

- Use CalypsoExamples/GeoModelTest program to dump as-built sensor positions
 - No alignment applied yet, but easy to use when available
- Determine the three unknown translations from the survey to the FASER coordinate system by requiring the average nominal and average survey positions for front sensors in Stations 1 – 3 to be equal.
 - Survey positions are adjusted accordingly to approximate the FASER coordinate system

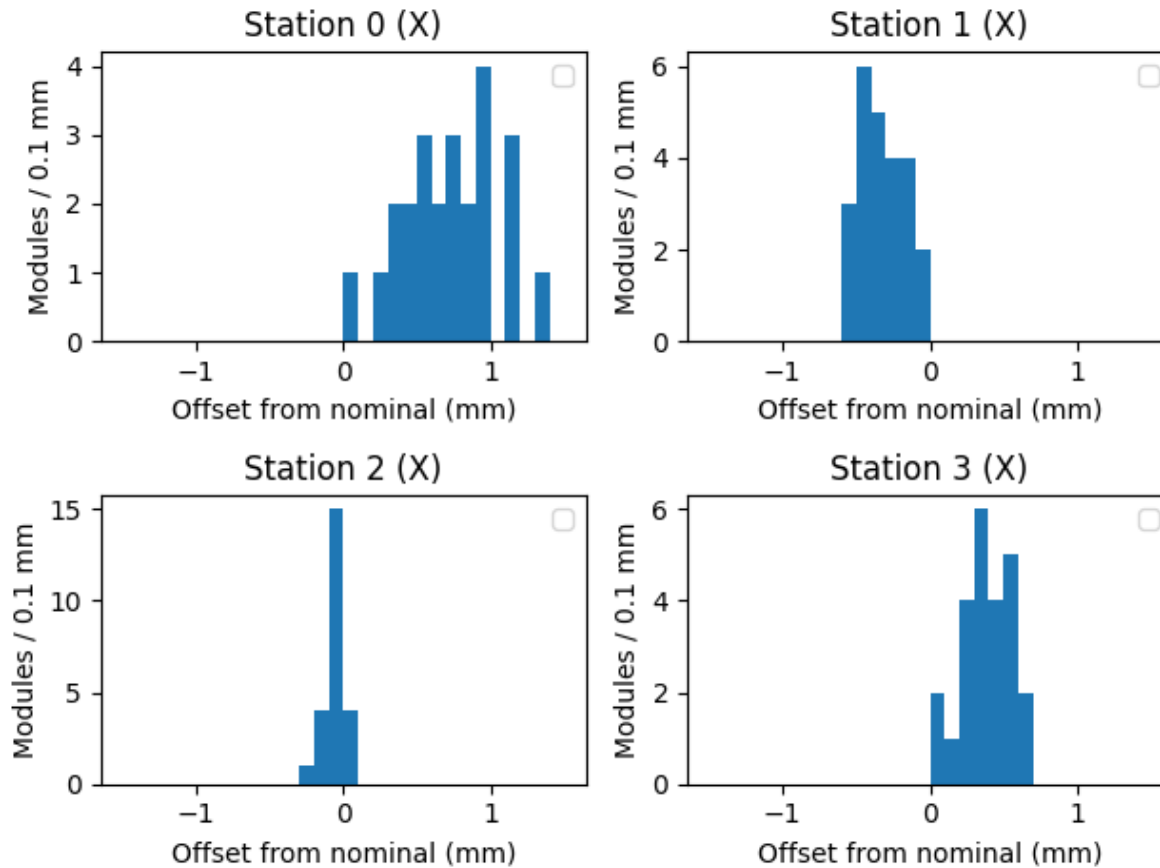
Average Offsets by Station

Tracker Station	X offset (mm)	Y offset (mm)	Z offset (mm)
0 (IFT)	0.727	-0.982	-27.772
1	-0.322	-0.095	0.121
2	-0.057	0.037	-0.037
3	0.379	0.058	-0.084

Standard Deviation of Offsets by Station

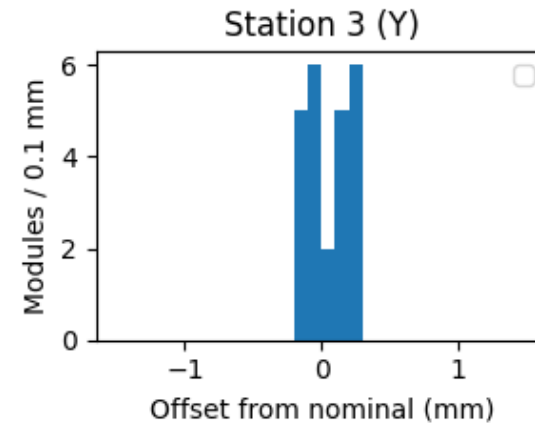
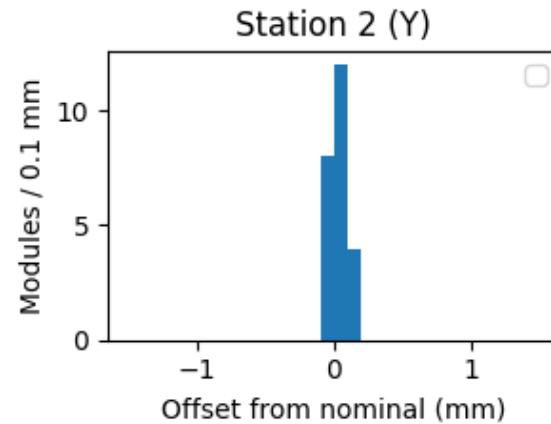
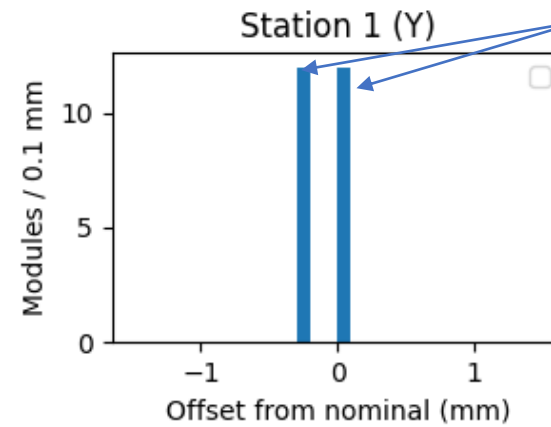
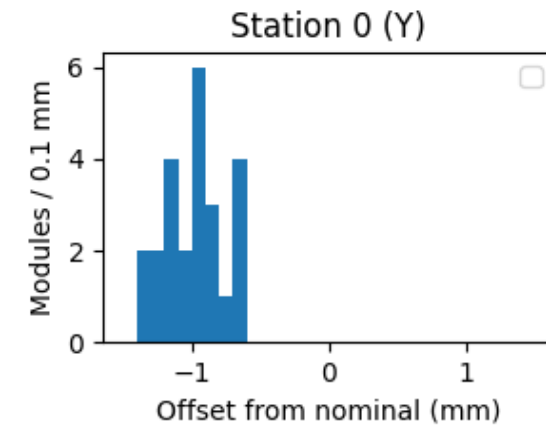
Tracker Station	X offset width (mm)	Y offset width (mm)	Z offset width (mm)
0 (IFT)	0.315	0.213	0.594
1	0.150	0.128	0.176
2	0.060	0.060	0.206
3	0.155	0.134	0.221

X Offsets



Stations 1 – 3 consistent with $350 \mu\text{rad}$ horizontal misalignment of support beam

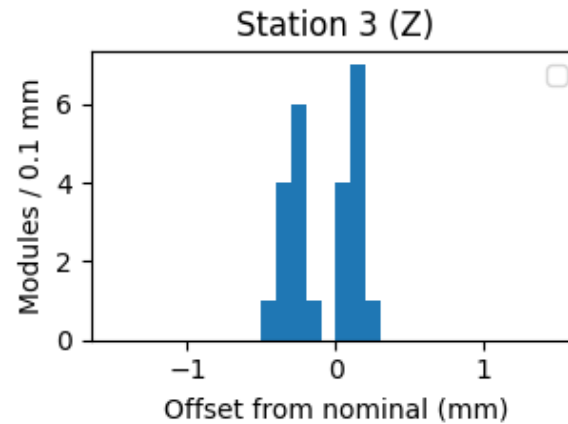
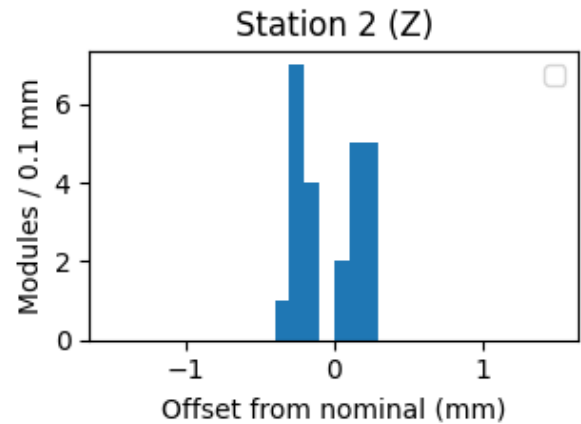
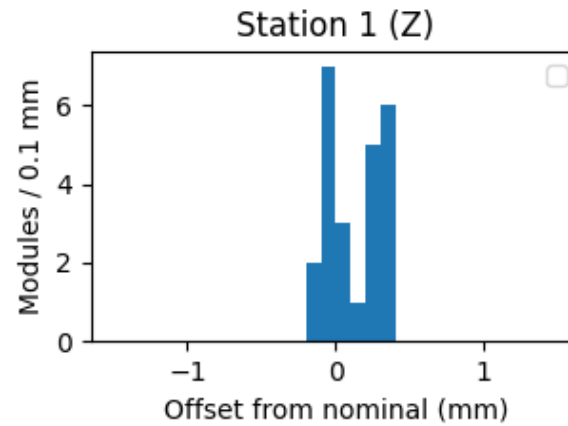
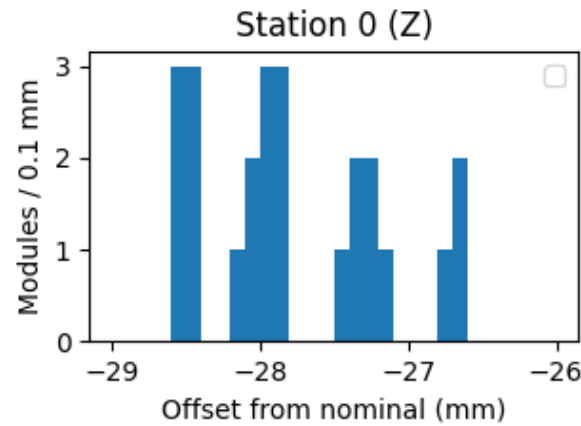
Y Offsets



No large offsets in the bending direction (apart from IFT...) is good!

Z Offsets

The IFT z shift is the only one we may want to account for in the nominal geometry



Bimodal z structure may be tied to wafer spacing issue

Mean Shifts by Layer

Mean layer shift from nominal (side = 0 only):

```
Layer..0.: Delta (x, y, z) = (.0.9667531209582592., ..-0.8338629886183355., ..-27.76697823415884.)
Layer..1.: Delta (x, y, z) = (.0.7550531702547865., ..-1.012504560299206., ..-27.785426685326456.)
Layer..2.: Delta (x, y, z) = (.0.4585192513611407., ..-1.100037257840885., ..-27.76415029087576.)
Layer..3.: Delta (x, y, z) = (.0.-0.30816863460421473., ..-0.08999266809102124., ..0.11269286539991308.)
Layer..4.: Delta (x, y, z) = (.0.-0.3291558805981598., ..-0.09403579372326121., ..0.13379340309479204.)
Layer..5.: Delta (x, y, z) = (.0.-0.3300890481152656., ..-0.10052884613507018., ..0.1163832240662046.)
Layer..6.: Delta (x, y, z) = (.0.-0.034812097241515794., ..0.05770992818386844., ..-0.0385857465991819.)
Layer..7.: Delta (x, y, z) = (.0.-0.05722013478983712., ..0.0440400335686455., ..-0.025114072848879232.)
Layer..8.: Delta (x, y, z) = (.0.-0.07793138066855931., ..0.009074111082156833., ..-0.046349153454116276.)
Layer..9.: Delta (x, y, z) = (.0.0.39666139814204016., ..0.05517815388198244., ..-0.09257975587377132.)
Layer..10.: Delta (x, y, z) = (.0.0.36871607015664765., ..0.07881538782588748., ..-0.0983307328406795.)
Layer..11.: Delta (x, y, z) = (.0.0.3719997077188921., ..0.0397396934073484., ..-0.06191003093516656.)
```

Intra-station shifts look very small